

Algorithmic Data Science & Asset Allocation

Instructor:	Dr. Aris Thorne	Term:	Summer/Fall 2026
Email:	athorne@univ.edu	Dates:	May 27 – Sept 15, 2026
Office:	Caruth Hall 412	Office Hours:	Friday 9:00 AM – 12:00 PM

COURSE DESCRIPTION

Introduces quantitative systems and optimization techniques used in automated wealth management. Explores efficient frontier calculations, data processing via Python/Excel, and portfolio asset selection algorithms.

GRADING POLICY & WEIGHTS

Assessment Activity	Weight
In-Class Live-Coding Participation	5%
Algorithmic Homework Problem Sets	20%
Data Science Simulation Labs	25%
Midterm Optimization Exam	25%
Automated Asset Allocation Capstone	25%

COURSE SCHEDULE

Date / Session	Topics, Assignments & Deadlines
May 29 (Fri)	Algorithmic Frameworks for Multi-Asset Universes
June 5 (Fri)	Module 1: Processing Financial Time-Series Data Matrices
June 12 (Fri)	Homework 1 Due: Covariance Matrix Generation
June 19 (Fri)	Lab 1 Due: Algorithmic Tracking of Portfolio Beta
June 26 (Fri)	Module 2: Markowitz Mean-Variance Optimization Frameworks
July 3 (Fri)	Module 2: Plotting the Efficient Frontier via Mathematical Solvers
July 10 (Fri)	MIDTERM OPTIMIZATION EXAM (Modules 1–2)
July 17 (Fri)	Module 3: Risk Parity Models & Equal Risk Contribution Strategies

Date / Session	Topics, Assignments & Deadlines
July 24 (Fri)	Homework 2 Due: Marginal Risk Contribution Analysis
July 31 (Fri)	Lab 2 Due: Risk Parity Code Integration
August 7 (Fri)	Module 4: Factor Investing Models & Multi-Factor Allocation Systems
August 14 (Fri)	Module 4: Backtesting Portfolio Allocations Against Historical Records
August 21 (Fri)	Homework 3 Due: Transaction Cost & Slippage Corrections
August 28 (Fri)	Lab 3 Due: Multi-Factor Strategy Robustness Assessment
September 4 (Fri)	Capstone Model Validations & Stress Testing Execution
September 11 (Fri)	Program Performance Reviews and Code Documentation Audits
September 15 (Tue)	AUTOMATED PORTFOLIO ALGORITHM SUBMISSION (11:59 PM)